



Yashika Bansal

B.Tech CSE (Information Security) · Full-Stack Developer · Cybersecurity
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PROFILE

Security-focused Full-Stack Developer and 2nd-year B.Tech student at VIT Vellore (CSE — Information Security, CGPA 8.5). I architect and ship production-ready applications — from AES-256 encrypted password vaults and multi-layer 2FA authentication systems, to 1,700-line zero-framework SPAs. Two internships completed, six live-deployed projects across cybersecurity and web development, with a consistent record of building complex systems independently.

EDUCATION

Vellore Institute of Technology (VIT), Vellore
B.Tech — Computer Science & Engineering (Information Security)

2024 – 2028
CGPA: 8.5 / 10

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), C, C++, Java, HTML5, CSS3, R
Frameworks & Libraries: React.js, Node.js, Express.js, Flask, Jinja2, Chart.js
Security & Cryptography: AES-256, bcrypt, XOR Cipher, PBKDF2, SHA-256, TOTP 2FA, OTP, SVG CAPTCHA
Databases: MongoDB, SQL, LocalStorage (client-side persistence)
Tools & Platforms: Git, GitHub, VS Code, Render, GitHub Pages, MATLAB, RStudio
UI/UX Engineering: Responsive Design, CSS Animations, Canvas API, Intersection Observer API, Dark/Light Mode

INTERNSHIPS

SAM AI Technologies — Cyber Security Intern May 2026 – Jun 2026 · Remote

- Architected and deployed 3 production-grade cybersecurity applications — CyberVault v3, Vault Auth System, and SecureVault XOR Encryption — all live on Render and GitHub Pages.
- Engineered multi-layer security infrastructure: AES-256 encryption, bcrypt hashing, TOTP 2FA, real-time entropy analysis, and layered XOR+AES file encryption with PBKDF2 key derivation.
- Implemented admin dashboards, SVG CAPTCHA generation, intelligent 3-attempt lockout policy, and immutable security audit logs tracking every authentication event end-to-end.

Hex Softwares Pvt. Ltd. — Web Development Intern Apr 2026 – May 2026 · Remote

- Conceived, designed, and shipped 3 live web applications — IRONFORGE Gym Platform, Bibliotheca Digital Library, and Personal Portfolio — entirely from scratch, zero frameworks.
- IRONFORGE: a 1,700-line zero-dependency SPA with 18+ interactive modules — BMI/TDEE calculators, Fitness RPG with XP levelling, AI chatbot, HIIT timers, and Chart.js progress dashboards.
- Bibliotheca: full-featured digital library — 50+ catalogue, borrow/return workflows, reading tracker, wishlist, peer reviews, Canvas API animated UI, and complete client-side data persistence.

PROJECTS (1 / 2)

CyberVault v3 — Password Security Platform Python · Flask · AES-256 · bcrypt · TOTP 2FA · JavaScript

- Full-stack password vault with AES-256 encryption, bcrypt hashing, and TOTP 2FA — production-grade credential security featuring a real-time entropy-based password strength analysis engine.
- Cyberpunk-aesthetic responsive UI with persistent secure storage architecture; deployed and live on Render.

↗ [cyberVault_v3](#) 📄 [Live Demo](#)

Vault Auth System — Enterprise Authentication Platform Python · Flask · bcrypt · Jinja2 · HTML · CSS · JS

- Multi-page enterprise-grade auth platform: OTP 2FA, SVG CAPTCHA, 3-attempt account lockout, admin user management — full Flask + Jinja2 MVC architecture, deployed on Render.
- Comprehensive immutable audit logs capturing every authentication event; hardened against brute-force and session-based attacks.

↗ [secure_login_system](#) 📄 [Live Demo](#)

SecureVault — XOR & AES File Encryption System Python · JavaScript · AES-256 · XOR Cipher · SHA-256 · PBKDF2

- File encryption/decryption tool with layered security — XOR cipher + AES-256 + PBKDF2 key derivation — SHA-256 hash integrity verification, and real-time password strength analysis.
- Paired with a 3-page interactive cryptography education site explaining the protocols powering the tool.

↗ [securevault-xor-encryption](#) 📄 [Live Demo](#)

PROJECTS (2 / 2)

IRONFORGE — Full-Feature Gym & Fitness SPA *HTML5 · CSS3 · Vanilla JavaScript · Chart.js · Zero Dependencies*

- ▶ 1,700-line zero-dependency SPA with 18+ modules: BMI/TDEE/calorie calculators, Fitness RPG with XP levelling, AI chatbot, HIIT/Tabata timers, 1RM calculator, and reaction speed game — no frameworks whatsoever.
- ▶ Dark/light mode, Chart.js progress dashboards, real-time class booking, supplement store with full cart logic — engineered entirely in pure HTML, CSS, and vanilla JavaScript.

↗ [HexSoftwares Gym](#) [Live Demo](#)

Bibliotheca — Digital Library Web App *HTML5 · CSS3 · JavaScript ES6+ · Canvas API · LocalStorage*

- ▶ 50+ book catalogue, borrow/return workflows, reading progress tracker, wishlist, peer reviews, and frontend user auth — all persisted client-side via LocalStorage with zero backend dependency.
- ▶ Dark Academia themed UI with Canvas API animated bookshelf background, CSS Grid/Flexbox responsive layout, and fluid scroll transitions throughout.

↗ [HexSoftwares-Library-Bibliotheca](#) [Live Demo](#)

AI Chef Cookbook — Hands-Free Voice & Gesture Dashboard *HTML · CSS · JavaScript · Web Speech API · Gesture Recognition*

- ▶ Touchless kitchen interface driven entirely by voice commands and hand gestures — eliminating all physical contact for maximum hygiene compliance in professional food preparation environments.
- ▶ Designed for professional kitchens where cross-contamination risk is high; enables complete recipe browsing and timer control without touching any device.

↗ [ai-chef-cookbook](#) [Live Demo](#)

Personal Portfolio *React.js · HTML · CSS · JavaScript · Intersection Observer API*

- ▶ Animated developer portfolio with Intersection Observer-triggered scroll reveals, smooth section transitions, and a fully responsive layout showcasing projects, skills, and internship work.
- ▶ Built with React.js for component-based architecture; features smooth parallax effects, animated skill bars, and an interactive project card gallery.

↗ [HexSoftwares-Personalportfolio](#) [Live Demo](#)

CERTIFICATIONS & EVENTS

NPTEL Elite — Human Computer Interaction · IIT Madras / SWAYAM, IIIT Delhi · Jan–Apr 2026 · 4 Credits · **Score: 80% (Assignments: 25/25)**

VIT Yantra Reboot 3.0 Hackathon · IEEE Robotics & Automation Society, VIT Vellore · Certificate of Participation

LANGUAGES

English (Professional Proficiency) **Hindi** (Native) **Spanish** (Conversational)